Alternative Algorithms for Social Media News Feeds: Audience Diversity and Stated Preferences-Based Algorithms

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Research Motivation

- Recommender systems prioritize content based on engagement signals, and low-quality content (e.g., polarizing, divisive) grabs attention quickly and drives high engagement → Feedback loops that may go against users' long-term interest in high-quality news feed.
- **RQ.** How can we redesign ranking algorithms to foster a healthier online information environment and enhance user agency?

Objective

Develop and evaluate two alternative (re)ranking methods that aim to address the limitations of engagement-driven algorithms: Audience Diversity (AD) and Stated Preference (SP) algorithms

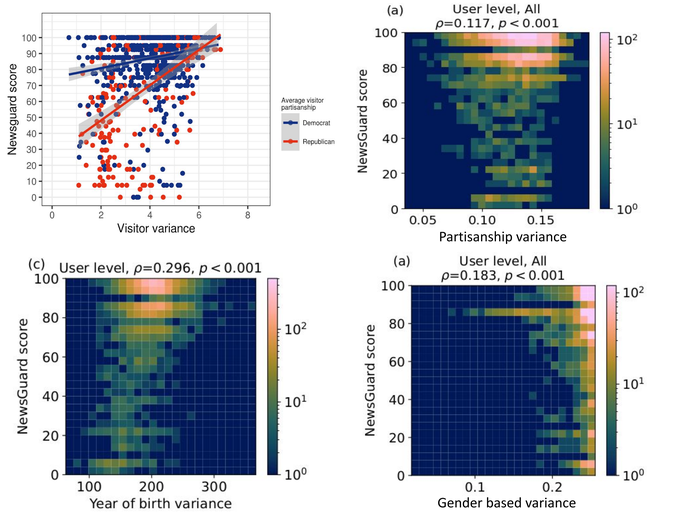
Audience Diversity Algorithm

Architecture

- Recommend news sources that generate high engagement and also attract a diverse set of audiences.
- *Audience Diversity (AD)*: The diversity in the ideological slants, age and gender of the audience that a news source has managed to attract.
- Implement AD as a system wide constraint, ensuring all users are exposed to sources with heterogeneous audience

Results

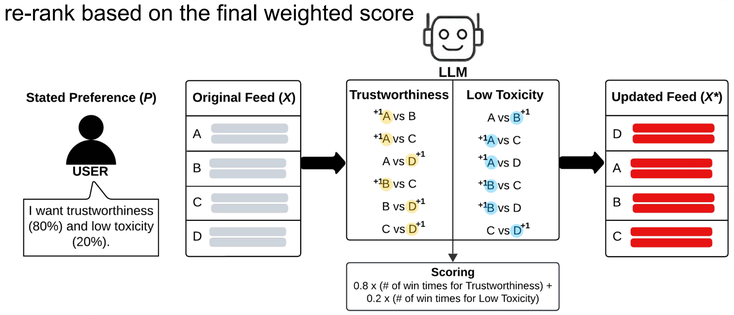
- Partisanship diversity calculated through clickstream data generated by YouGov Pulse Panel is positively correlated with journalistic ratings of news sources (top left figure) (Bhadani et al. 2022).
- Partisanship diversity, age based diversity and gender based diversity calculated through social media sharing data (obtained from DomainDemo) is also positively correlated with journalistic ratings.
- For both clickstream data and social media sharing data, popularity of news sources is not correlated with journalistic ratings.



Stated Preference-Based Re-Ranking Algorithm

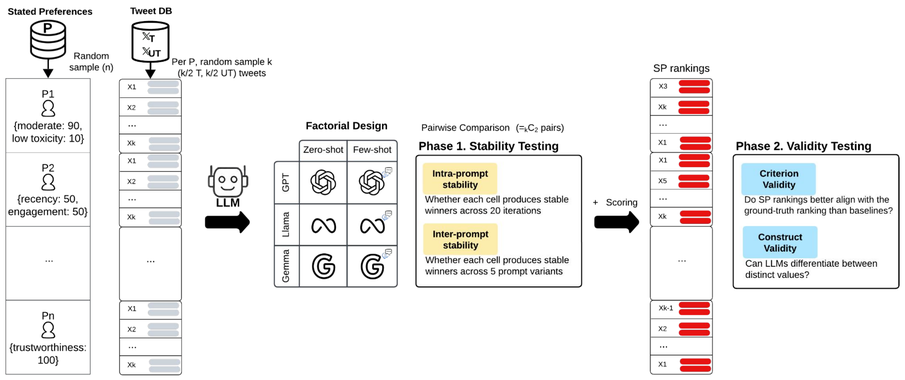
Architecture

- More user agency → Lets users rewrite their timeline by describing their desired qualities and values (i.e., stated preferences) in natural language.
- Leverages LLMs to perform pairwise comparisons of given content paris: assess which content better align with user-defined value priorities (i.e., winner item gets +1), and re-rank based on the final weighted score



Evaluation Method

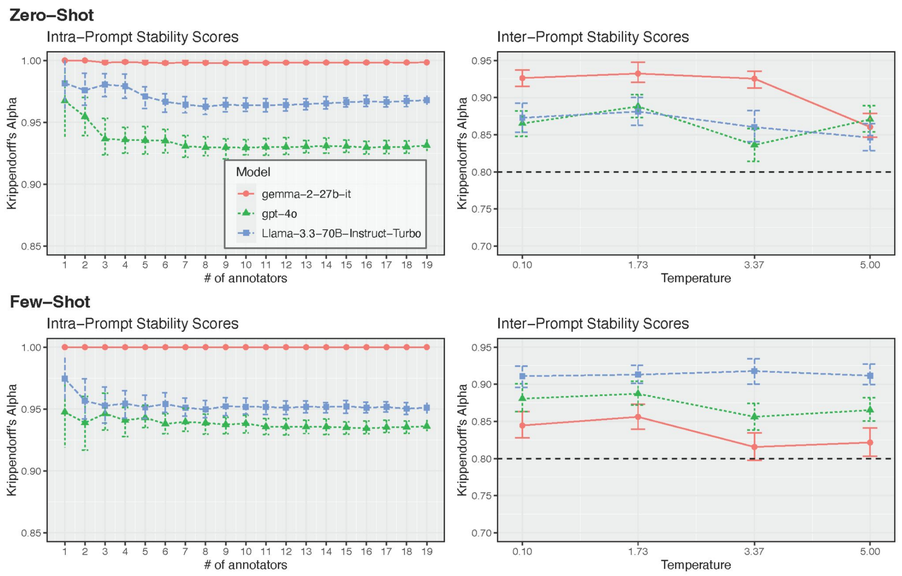
- 3 x 2 factorial design by model (GPT-4o, Llama-3.3-70B-Instruct-Turbo, Gemma-2-27b-it) x prompting strategy (Zero-shot, Few-shot)
- **Stability**: Does SP algorithm produce stable pairwise content comparisons?
- **Validity**: Does it yield rankings aligned with users' values, and behave as intended?



Results

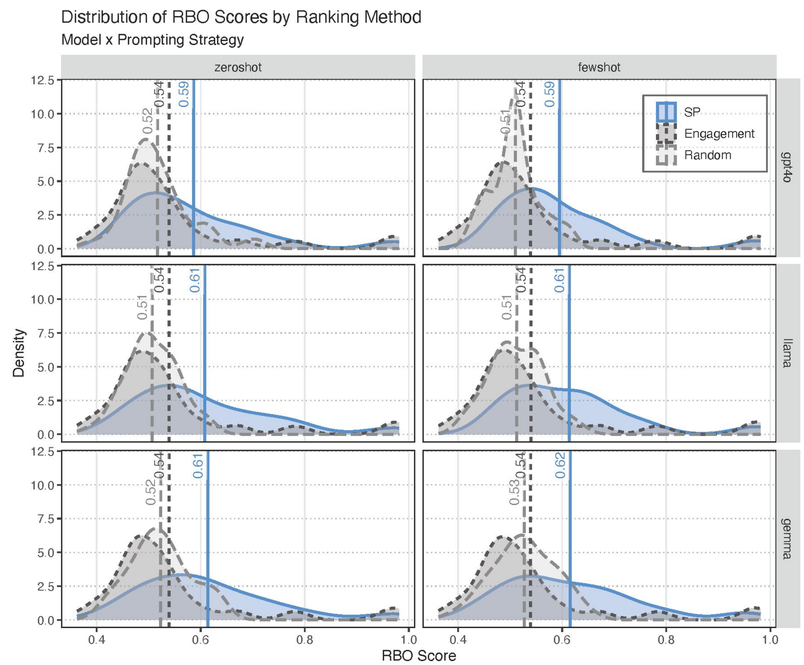
1. Stability Testing

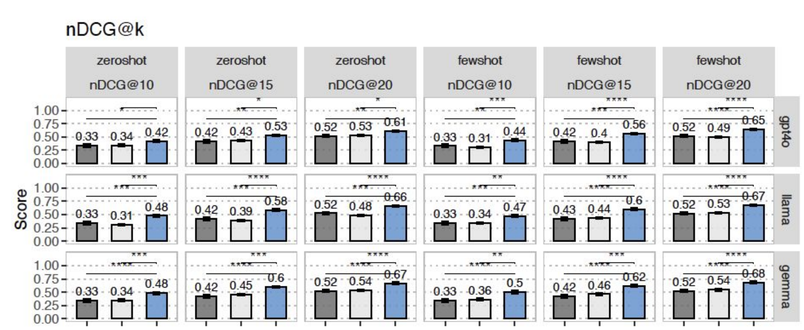
- Intra-Prompt Stability: stable across multiple iterations (=# of annotators)
- Inter-Prompt Stability: stable across semantically similar prompts



2. Validity Testing

- Criterion Validity: SP rankings were more similar to the ground truth ranking compared to baselines (=higher RBO), and outperformed them in terms of information retrieval metrics (e.g., Precision@k, Recall@k, nDCG@k)
- Construct Validity: *(Ongoing)*





Implications

- **AD:** Designed to enhance the trustworthiness of news feeds by using audience diversity of news sources as a signal of their quality.
- **SP:** Proof-of-concept that LLMs can realign a news feed with users' stated preferences in a stable and valid manner, pointing toward algorithms that serve what people say they want rather than only what they engage with.

Going forward

- More tests on SP + a lightweight demo for user interaction
- Currently preparing an RCT to examine how these alternative ranking algorithms affect user behavior and attitudes

References

Bhadani *et al*. Political audience diversity and news reliability in algorithmic ranking. *Nature Human Behaviour* **6**, 495–505 (2022).

Acknowledgements

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